

Santiago Lopez Carreon

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<https://github.com/SantiagoELC-Git>

SUMMARY OF QUALIFICATIONS

- Strong interpersonal skills developed through extracurriculars.
- Strong oral and written communication developed through coursework and extracurricular activities.
- Strong computer literacy with general software such as Office 360 and Jupyter developed through coursework, personal interest, and extracurriculars.
- Proficiency in Python developed through work experience, extracurriculars, and coursework. Proficiency with HTML, CSS, and JavaScript developed through work experience and personal projects.
- Proficiency in MATLAB through online course MATLAB on-ramp, with extra practice with MATLAB community problems.

WORK EXPERIENCE

Data Scientist/Machine Learning Co-op

Jan. 2026 – April. 2026

BBD Insights

Toronto, ON

- Contributing to backend development of risk model governance software, supporting risk model inventory and oversight workflows.
- Developed a functional REST backend API with Python using Flask library
- Developed production-ready systems for AI-driven document summarization using RAG and function-calling patterns with Python libraries (LangChain, google-genai), integrated via Flask; implemented exponential backoff and retry logic for robust API interactions.

VOLUNTEER EXPERIENCE

Vice-President of Finance

Jan. 2025 – Aug. 2025

University of Waterloo Undergraduate Physics Society

Waterloo, ON

- Oversaw financial operations of the undergraduate Physics Society, including budgeting, reimbursements, expense tracking, and collaboration with executives and governing boards to ensure transparent and strategic use of funds.
- Managed and reimbursed over \$3,500 in budgets for community events and office supplies, ensuring accurate financial tracking and compliance.
- Utilized spreadsheets and Google Forms to monitor expenses and streamline reporting for teams and the Science Society Board of Directors, which led to clear and fast retrieval of information and a decrease in mistakes from all executives.
- Collaborated with executives to secure funding for events that enhanced student engagement and ensured financial returns benefited the student community.

First Year Representative

Sep. 2024 – Dec. 2024

University of Waterloo Undergraduate Physics Society

Waterloo, ON

- Advocated for first-year Physics students by bridging communication with Society executives and promoting engagement in community events.
- Coordinated timetables and updates between students and executives.

Physics and Math Tutor

May. 2025 – Dec. 2025

University of Waterloo Physics Tutorial Centre

Waterloo, ON

- Delivered one-on-one help in core Mathematics and Physics concepts to engineering first years by taking advantage of metaphors and self-made diagrams to explain concepts
- Applied interpersonal skills to create a comfortable and engaging learning environment through friendly conversation and positive encouragement throughout the session

Volunteer Research Assistant

Nov. 2025 – April. 2026

University of Waterloo Theoretical Chemistry Department

Waterloo, ON

- Developed a testing process with libraries in juliacall and pytest, that can compare functions from Julia and python check that the output is the same

CERTIFICATES

Physiological First AID

Certificate #: 104973770

First AID and CPR/AED Level C

Certificate #: 105003987

TECHNICAL SKILLS

Languages: Python, MATLAB, LaTeX, Julia, HTML, CSS, JavaScript

Libraries: NumPy, Matplotlib, Pandas, SQLAlchemy

Frame Works: Flask, Pytest

Softwares: Office360, Google Sheets, Excel, Microsoft VS Code

LANGUAGES

Spanish: Passive bilingualism

EDUCATION

University of Waterloo

Sep. 2024 – April. 2029(Expected)

Bachelor of Science in Mathematical Physics

Waterloo, ON

- Student ID: 21122111

SCHOLARSHIPS AND AWARDS

University of Waterloo President Scholarship

Principal Award - St. Ignatius of Loyola Catholic Secondary Scholar